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RESEARCH ARTICLE

Improved Q-Learning-Based Motion Control for Basketball Intelligent Robots Under Multi-Sensor Data Fusion

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ABSTRACT With the continuous development of science and technology, the application of artificial intelligence in robot technology, especially in the progress of sports robot technology, has received great attention. This development provides new opportunities for the development of intelligent basketball robots. However, achieving high accuracy and stability in motion control remains a challenge in the research field. The motion control technology of basketball intelligent robots is crucial. The research proposes a study on enhancing Q-learning-based action control technology for basketball robots with multi-sensor data fusion to address the problem of inaccuracy and instability. The first step involves examining how multi-sensor data fusion technology can offer more accurate and comprehensive environmental information that can aid the robot in making more precise action decisions. The second area of study involves the development of enhanced algorithms for controlling basketball robots through improved Q-learning techniques. The goal is to increase the accuracy and stability of the robots' actions. The final focus is on creating a model for controlling the basketball robots' actions, which is based on the improved Q-learning algorithm. This model effectively increases the accuracy and stability of the robots' actions. The experiment was conducted on a computer equipped with Intel Core i9-11900K processor and NVIDIA GeForce RTX 3080 graphics card. MATLAB R2023a was used for system parameterization and simulation, while Webots R2023a software was used to simulate robot actions. The sensor adopted Hokuyo URG-04LX-UG01 laser rangefinder, and the robot model was KUKA youBot. At the same time, in order to improve the control performance of basketball intelligent robots, comparisons were made with existing methods such as PID control, trajectory planning, traditional control methods, multi-sensor fusion technology, and visual perception basketball shooting correction technology. The experimental results demonstrated that the enhanced Q-learning algorithm performed exceptionally well, achieving a recognition rate of 99.35%, a response speed of 67.85%, remarkable performance stability of 97.55%, and a low error rate of 7.65%. The improved Q-learning algorithm's efficiency and stability were confirmed in the application of robot control modeling. Compared to existing methods for controlling basketball robots' motion, the proposed method can handle environmental uncertainty more effectively and improve decision-making accuracy. The research has several advantages in robot motion control. It enhances the applicability of basketball intelligent robots in actual competitive sports and provides technical support for robot strategy formulation, rapid adaptation, and efficient execution in actual competitions. It serves as a crucial source for future studies on technology for controlling robot movements and the practical implementation of intelligent basketball robots.

INDEX TERMS Multi-sensor data fusion, Q-learning algorithm, motion control, basketball robot, DQN algorithm.

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I. INTRODUCTION

Motion control technology is essential in improving the performance of robot movements, particularly in sports

competitions like basketball where high precision and coordination are increasingly demanded. While existing motion control technologies have enhanced the basic motion performance of robots, they still have limitations in handling dynamic decision-making and precise action execution in high-speed and complex sports activities. The research aims to improve the motion ability of robots in high dynamic sports activities through advanced machine learning algorithms and data fusion technology. This will promote the widespread application and development of intelligent sports robots in practice. Motion control technology can enhance the basic functions of robots such as walking, running, and jumping, while improving the accuracy and coordination of these actions. By providing robots with more natural and smooth motion capabilities, motion control technology enables them to collaborate and perform tasks more effectively in various fields, such as industrial automation, the service industry, and healthcare [1]. Additionally, robots have the potential to excel in motion control (MC) scenarios that require high precision and coordination. One product of this cross-fertilization is the intelligent basketball robot (IBR), which has gained attention in recent years [2]. MC technology utilizes robots for high-speed and dynamic sports activities. The robots require not only high physical abilities to imitate human athletes but also extremely fast decision-making and reaction abilities to execute precise actions in complex competitive environments [3]. Intelligent robots designed for basketball require the ability to process sensor data from various sources and make quick judgments. This necessitates the use of precise data fusion technology and learning algorithms [4]. Q-Learning is a reinforcement learning algorithm that enables the robot to gradually find the optimal strategy through continuous trial-and-error learning. Multi-sensor data fusion (MSDF) technology can fuse information from different sensors, improving the robot's perceptual ability and making it better able to deal with complex environments [5]. However, the basketball intelligent robot cannot accurately simulate the complex actions of basketball players, such as shooting, dribbling, and steals in real-time when faced with data from different sensors. Additionally, it may struggle to make quick decisions in dynamic and unpredictable competition environments. This study proposes an MC-IBR technique based on improved Q-Learning under MSDF. The goal is to improve the accuracy, response speed, and stability of the robot while minimizing errors. The proposed technique aims to improve the robot's perceptual ability and provide support for its operation in complex environments. This study applies an improved Q-learning algorithm to the action control of basketball intelligent robots, taking into account the dynamic characteristics of basketball and the motion constraints of robots. The robot's dynamic perception ability of the basketball game environment is enhanced through MSDF, enabling it to make more accurate and quicker action decisions and executions. The present study can improve the level of MC-IBR and lay the foundation for its application in actual

competitions, which is of great significance in promoting the development of IBR.

The study contains five main parts. The first part is the introduction, which describes the impact of IBR on the field of life and sports training in the context of the rapid development of artificial intelligence and robotics. The second part is the literature review, which summarizes the applications of IBR in various fields and the current status of research. The third section is an in-depth study of the improved QLA under MSDF. The article consists of five sections. The first section discusses the application of MSDF technology. The second section presents the improved basketball robot motion control (BRMC) algorithm for enhanced Q-Learning. The development of the BRMC model using the enhanced QLA is covered in the third section. The fourth section evaluates the performance of MC-IBR through testing. The final section provides a summary and outlook of the research methodology and results.

II. RELATED WORKS

With the rapid development of artificial intelligence technology, the use of improved Q-Learning-based techniques under MSDF to realize MC-IBR has attracted widespread attention, and many scholars have already had quite a lot of research results. According to Zhang and colleagues, coherent beam combination was a useful method for overcoming the power constraints of single fiber lasers. QLA was one type of reinforcement learning algorithm that can be used for this purpose. The outcomes demonstrated the good stability and ease of debugging of this QLA [6]. Castaldo et al. proposed an algorithm to solve the optimal population transfer problem for molecules affected by laser pulses. The algorithm was simulated on a quantum computer as a molecular wave function and combined with iterations of a machine learning algorithm to form the optimal pulse. The performance was evaluated on a noisy medium-scale quantum device and compared with the performance of standard evolutionary algorithms. According to the findings, the hybrid algorithm offered a higher degree of algorithmic performance and performed better in terms of optimization than the downhill simplex approach [7]. Suat et al. proposed a mechanism based on reinforcement learning to improve the health status of patients. The mechanism employed an intervention selection model and an opportunity moment recognition model in order to personalize interventions in terms of timing, frequency and preferred type. Preferences for specific intervention types and attitudes toward target behaviors were analyzed through simulation experiments. The outcomes revealed that the method produced better results than standard reinforcement learning algorithms and effectively improved the health status of patients [8]. A cohort of patients with external metastatic long bone disease was given accurate survival predictions using machine learning algorithms created by Skalitzy et al. in response to the problem of optimizing survival prediction for the treatment of metastatic long bone

disease. In terms of discrimination, calibration, overall performance, and decision curve analysis, the study's findings demonstrated that the machine learning algorithm continued to operate well [9]. He et al. proposed a variational quantum compilation algorithm based on reinforcement learning. The algorithm selected quantum gates sequentially from the native gate alphabet and the quantum bits they act on by means of dual Q-Learning, ϵ -greedy exploration strategy and empirical playback. The results demonstrated that the method can achieve accurate compilation with fewer quantum gates, reduce quantum algorithmic errors caused by de-coherence processes and gate noise, and improve the efficiency and accuracy of running quantum algorithms on noisy medium-sized quantum devices [10].

Improved Q-Learning can fuse complex basketball motion strategies and complex sensor data to optimize and improve the robot's MC through machine learning methods to improve its performance in basketball, which has been widely noticed and studied by many scholars by virtue of its many advantages. Using variational Bayesian inference and a sparse Gaussian process-based prior, H. Sasaki devised a nonparametric policy search approach with many optimal actions. The results indicated that the algorithm can effectively achieve learning optimization and improve the performance of reinforcement learning for policy search [11]. For the robot linked stability problem, A Usova suggested a control design framework that permits stabilization but not necessarily passive robot-environment interaction. The results showed that the framework can be used in conjunction with tracking control algorithms for arbitrary robots [12]. To ensure that the target follows its reference trajectory using only collisions as discrete inputs, Li et al. proposed a trajectory tracking method that considers discontinuities in the rib drop system. The tracking approach was based on the reference diffusion error and included prediction, re-planning, and tracking of the target system and tracking system. The study's findings demonstrate the viability and efficacy of the suggested control technique [13]. Taylor et al. provided an overview of active learning in robotics, emphasizing how it can be adapted to the needs of lifelong learning systems. The control-oriented challenge was to select appropriate target measures with limited understanding of the environment or the robot itself. The findings indicated that robotic active learning systems need to consider learning task categories, measures of information content, and algorithms for action plan generation [14]. Hoang et al. proposed an augmented trajectory tracking method for nonlinear system control. The method combined system dynamics with an extended quadratic port-Hamilton model without the use of input and state coordinate transformations in order to investigate the control benefits. The results indicated that the control system improved the closed-loop performance and robustness, validating the effectiveness of the method [15].

In summary, the improved Q-Learning-based robot MC technique has a significant impact on existing basketball

robot control technology. The combination of MSDF and the improved Q-Learning method is crucial for the advancement of robot MC technology. The MC-IBR technique proposed in this study can learn from complex input data and optimize the output based on feedback. The goal is to enhance the robot's ability to adapt to the intricacies of basketball, ultimately advancing the robotics industry as a whole. Despite the unique advantages of improved Q-Learning-based MC techniques for robots under MSDF, there are still unsolved challenges. Further research is needed to address issues such as handling high-dimensional sensing data more efficiently and making faster decisions in real-time environments.

III. MC-IBR TECHNOLOGY WITH IMPROVED Q-LEARNING BASED ON MULTI-SENSOR DATA FUSION

To explore the application of MSDF technology, firstly, the characteristics of various types of sensors and fusion algorithms are analyzed in detail to provide theoretical support for the perception ability of basketball robots. Secondly, MC technology with improved Q-Learning is explored to improve the decision-making ability and MC efficiency of basketball robots with respect to their characteristics and needs. Finally, a BRMC model based on the improved QLA is constructed to provide an effective mathematical model for BRMC. This study comprehensively investigates the perception and MC techniques of IBR through the fusion of multi-sensor data and the optimization of QLA by using DNQ network, expecting to provide a powerful theory and a new idea for the development of basketball robotics.

A. MULTI-SENSOR DATA FUSION TECHNOLOGY

Multi-sensor information fusion (MSIF) technology plays an important role in areas such as smart manufacturing and condition monitoring due to its wide observation capability, high confidence information, real-time measurements and low cost advantages [16]. The study intends to use this technique for robot motion parameter calibration to improve positioning accuracy, which involves data pre-processing and analysis of information fusion algorithms. MSIF is the technique of obtaining accurate target information using multiple sensor data [17]. In MC-IBR technique, the study utilizes improved Q-Learning for optimization. In practice, by pre-processing and fusing data from different sources with MSIF technology, the robot's state and environment information can be acquired more accurately and in real time to improve the motion performance and adaptability [18]. The MSIF process is shown in Figure 1.

The basic process of MSIF technology consists of five main steps: raw data acquisition, measurement data pre-processing, data correlation, data decision making and information fusion [19], [20]. Target state information is collected by multiple sensors, and these data are temporally and spatially aligned. Then the correspondence between data and targets is established, and data decision-making is carried

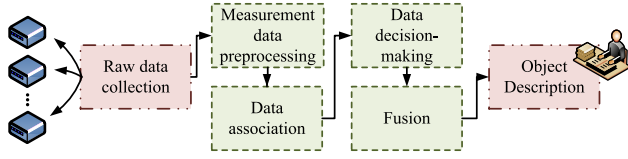


FIGURE 1. Multi-sensor information fusion process.

out to select the optimal target in the presence of multiple targets in space. Finally, the multi-sensor measurement data are synthesized and processed by certain algorithms to get the fusion results closest to the real state. Figure 2 displays the MSIF algorithm's schematic diagram.

Sensor information fusion is a dynamic, nonlinear process that requires processing information from a variety of sensors and resolving conflicts and inconsistencies between the information [21]. Through multi-level fusion of space, time and features, it is possible to realize omni-directional and multi-level perception of the target [22]. Among them, spatial fusion is mainly achieved by optimizing the spatial layout of multiple sensors to achieve omni-directional perception of the target. Temporal fusion is to realize continuous tracking of the target by fusing the information acquired by the same sensor at different times. Feature fusion, on the other hand, realizes high-precision recognition of the target by extracting the common features in the information of each sensor. The equation for its recognition degree is shown in equation (1).

$$n_{ij} = f(|x_i - x_j|) \quad (1)$$

In equation (1), n_{ij} is the degree of certainty, and x_i and x_j are the measured data of the two types of sensors after spatio-temporal alignment. In order to realize the information fusion of multiple sensors, the fusion information of the integrated correctness is shown in equation (2).

$$\eta = Pa \quad (2)$$

In equation (2), η is the integrated correctness of the data, P is the real symmetric matrix, and a is the column vector. Assuming that the measurement system consists of two sensors, since the truth value to be estimated is a constant, it is necessary to estimate the truth value of the object parameter by the average value of the sensor's historical measurement data, which is calculated as the relationship shown in equation (3).

$$\bar{X}_i(m) = \frac{1}{m} \sum_{k=1}^m x_k \quad (3)$$

In equation (3), m is the number of historical measurements of the sensor, and $\bar{X}_i(m)$ is the average value of historical measurements of the sensor. The premise of the optimal weighting factor is to first find the variance of each sensor, which can be based on the measurement values of each sensor [23]. Let the measurement errors of two mutually independent sensors

be e_i and e_j , respectively, then the measured values X_i and X_j , which are calculated as shown in equation (4).

$$\begin{cases} X_i = X + e_i \\ X_j = X + e_j \end{cases} \quad (4)$$

In equation (4), X_i and X_j are measured values and e_i and e_j are zero-mean Gaussian white noise. Then the two sensors variance, which is shown in equation (5).

$$\sigma_i^2 = R[e_i^2] \quad (5)$$

In equation (5), σ_i^2 is the variance of the two sensors. MSIF is an effective way of processing and parsing data and its mathematical expression is shown in equation (6).

$$\tau_i = k_0 l_i (\Delta \theta_i) \quad (6)$$

In equation (6), k_0 denotes the weight of each sensor data, l_i denotes the data detected by each sensor, and $\Delta \theta_i$ denotes the fused data. The action control based on improved Q-Learning can be characterized using the theory of statistics to provide theoretical basis and technical support for further research and application.

B. MOTION CONTROL ALGORITHM BASED ON DNQ NETWORK OPTIMIZED Q-LEARNING

In recent years, Q-Learning, as a model-free reinforcement learning algorithm, has been widely used in robot behavior control due to its good self-learning and self-adaptive performance [24]. However, the traditional QLA is inefficient in learning when facing complex, dynamic, and uncertain environments. It is difficult to meet the needs of basketball robots for fast learning and decision-making in real matches. Therefore, effectively describing and utilizing such complex algorithms becomes a pressing issue. Therefore, Q-Learning is utilized to learn and optimize each action of the robot to achieve a specific basketball skill. The Q-Learning process and DNQ network structure are schematically shown in Figure 3.

The improved QLA has specific learning and optimization properties, and the dynamic performance of this algorithm in BRMC is achieved through MSDF [25], [26]. Compared with traditional Q-learning methods, this algorithm can not only enhance the perception ability of basketball robots towards environmental states, but also effectively handle more complex state spaces and action spaces than traditional Q-learning methods. Firstly, the DNQ network effectively extracts visual features in complex environments through its deep convolutional neural network layers, while utilizing recurrent neural networks to partially capture dynamically changing time series information. Therefore, it enables the DNQ network to learn more comprehensive and detailed environmental state representations from multi-modal sensor data, such as visual sensors, inertial measurement units, and wheel speed sensors. Secondly, the DNQ network introduces an Experience Replay mechanism and a Target Network, which improves the efficiency of the algorithm in handling

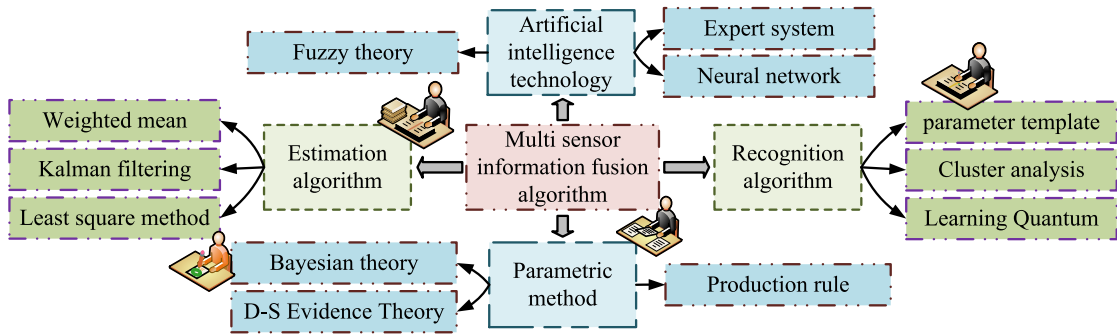


FIGURE 2. Multi-sensor information fusion algorithm diagram.

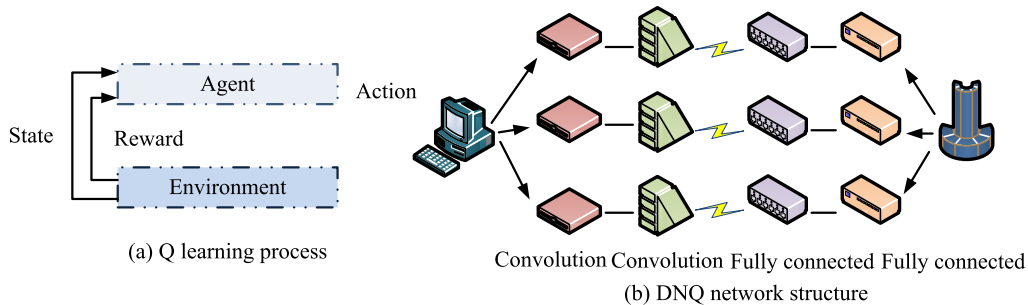


FIGURE 3. Q-Learning process and DNQ network structure diagram.

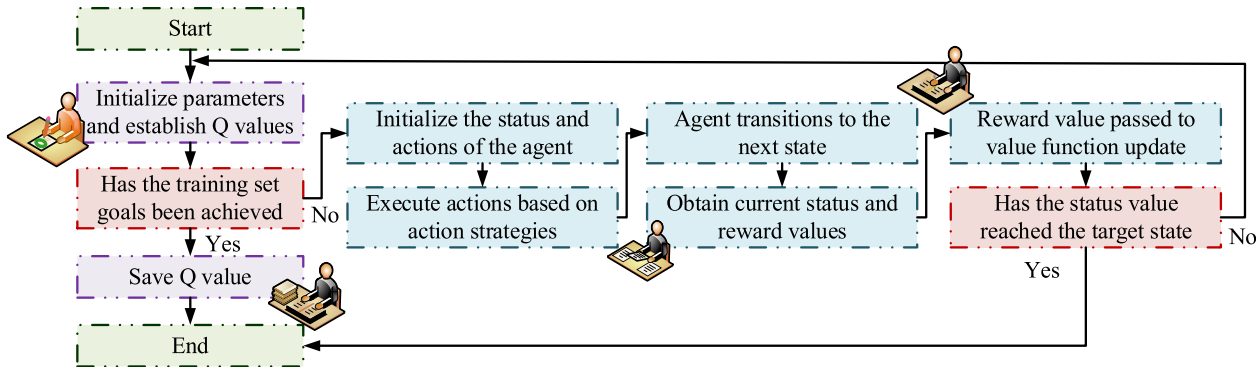


FIGURE 4. The motion control process is based on improved Q-Learning algorithm.

complex state space learning while ensuring stability during the learning process. In addition, the DNQ network efficiently encodes the vast state action space, which not only enhances the algorithm’s generalization and discrimination ability in high-dimensional decision spaces, but also enables basketball robots to achieve more precise and stable action control in complex and dynamic environments. The MC process based on the improved QLA is shown in Figure 4.

The improved QLA is able to find the optimal decision-making strategy in a large amount of data while maintaining stability and real-time performance [27]. A mechanism for adaptive learning rate adjustment has been

incorporated into the action control of basketball robots. This mechanism automatically adjusts the learning rate based on the algorithm’s convergence. Additionally, the robot can balance new actions and utilize the current optimal action by combining the ϵ -greedy strategy and upper limit confidence interval strategy. By mimicking the decision-making process and characteristics of this learning algorithm, control strategies and systems that can perform similar functions are designed. The improved QLA has a unique way of distributing and acquiring rewards during selection and execution, which is advantageous for designing intelligent systems and machine learning models. Its mathematical expression for

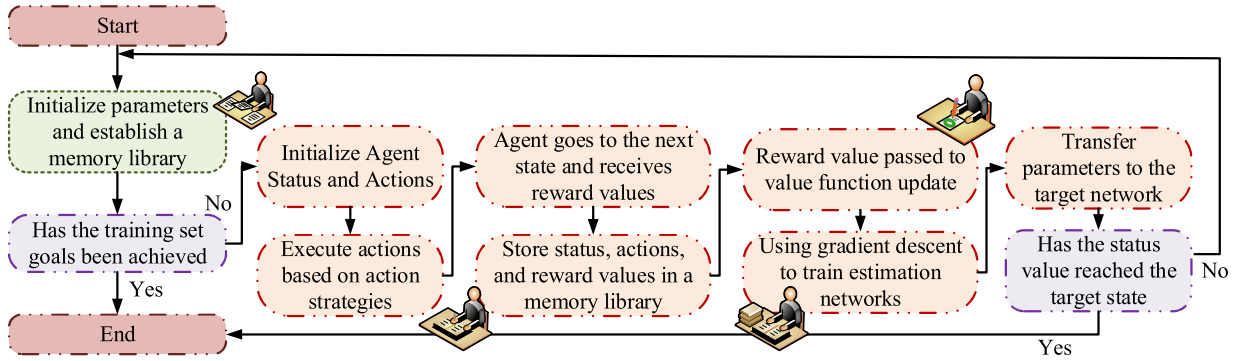


FIGURE 5. Motion control and decision process based on DNQ improved algorithm.

updating the Q-function is shown in equation (7).

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t) \right] \quad (7)$$

In equation (7), $Q(S, A)$ denotes the action state value, α denotes the step parameter, and γ denotes the discount factor. Moreover the mathematical expression of the Agent outputting the optimal policy is shown in equation (8).

$$P(a|s) = \frac{e^{\frac{Q(s,a)}{T}}}{\sum_{b \in A} e^{\frac{Q(s,b)}{T}}} \quad (8)$$

In equation (8), T denotes the exploration factor. P denotes the state transfer probability. The state direction of the basketball robot keeps changing, and for action stability, the reward is constant, while for action uncertainty, the reward changes as the state will change. Additionally, the MC and decision process based on DNQ improved algorithm is shown in Figure 5.

The state transfer matrix based on the improved QLA can be used to plan and control the movements of the basketball robot, as well as to analyze the decision-making performance for it. The mathematical expression of its network update is shown in equation (9).

$$\theta_{t+1} = \theta_t + \alpha \left[r + \gamma \max_{a'} Q(s', a'; \theta') - Q(s, a; \theta) \right] \nabla Q(s, a; \theta) \quad (9)$$

In equation (9), $Q(s, a; \theta)$ denotes the real-time updated value of the network. $Q(s', a'; \theta')$ denotes that the updated network will synchronize the parameters once in a while. The BRMC strategy based on the improved QLA realizes flexible, efficient and precise control. It provides a new perspective and technical path for the design and manufacture of future robots.

C. CONSTRUCTION OF BASKETBALL ROBOT MOTION CONTROL MODEL BASED ON IMPROVED Q-LEARNING ALGORITHM

BRMC is a crucial research area in the field of artificial intelligence. Its objective is to enable robots to autonomously play basketball [28]. However, current BRMC methods have limitations, such as poor adaptability to changes in the field environment and learning effects, which affect decision-making accuracy. To address these limitations, new mechanisms have been introduced to enhance the basketball robot's MC capability, making it more flexible and intelligent during game-play. The structure of the BRMC system based on the improved QLA is shown in Figure 6.

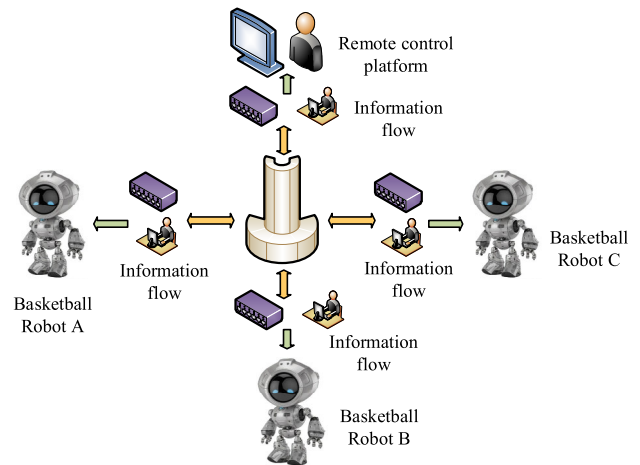


FIGURE 6. Motion control system structure of basketball robot based on improved Q learning algorithm.

The BRMC system structure, contains three main links: sensing, decision making and execution [29]. The sensing link will acquire real-time information on the basketball court through various sensors, and convert this information into state representations as the basis for learning and decision-making. The decision-making session finds the optimal behavioral strategy through continuous exploration and learning. The execution link then converts the behavioral strategies

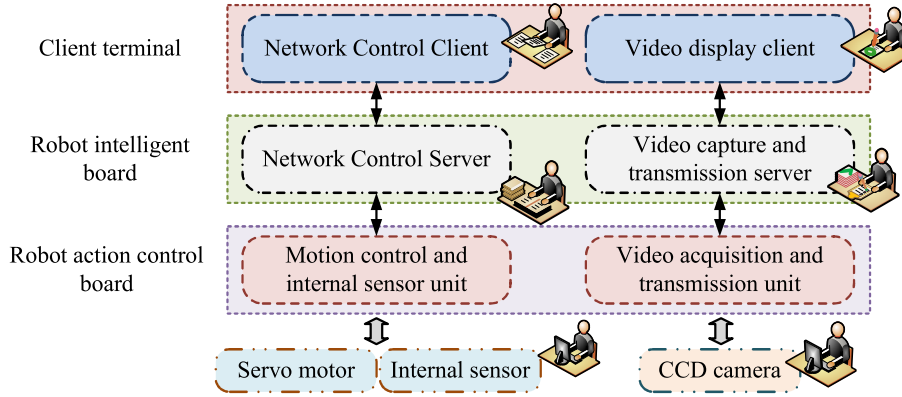


FIGURE 7. Hardware composition of the closed-loop control system for pneumatic robotic arms.

generated by the decision-making link into specific robot actions. The perception module integrates various sensors, including visual sensors, inertial measurement units, ultrasonic sensors, and touch sensors, to obtain comprehensive information on the basketball court. Inertial measurement units provide real-time data on acceleration and angular velocity for robots, while visual sensors capture visual information between robots and the environment. Torque sensors measure the force and torque of robot joints. After analyzing the data with pre-processing and information fusion algorithms, more comprehensive state information can be provided. Q values need to be iteratively updated if they want to reach a stable convergence state, and their mathematical expressions are shown in equation (10).

$$\Delta Q = r_1 + \gamma \max_{a_t \in A} Q_{t+1}(s_{t+1}, a_t) - Q_t(s_t, a_t) \quad (10)$$

In equation (10), r_1 denotes the samples in the storage learning process. In this case, the DNQ algorithm converts the updated Q-value to update the parameters of the network, using the gradient descent method to update the neural network parameters θ , whose mathematical expression is shown in equation (11).

$$\begin{aligned} \theta_{t+1} &= \theta_t + \alpha \left[r + \gamma \max_{a'} Q(s', a'; \theta') - Q(s, a; \theta) \right] \nabla Q(s, a; \theta) \end{aligned} \quad (11)$$

In equation (11), $r + \gamma \max_{a'} Q(s', a'; \theta')$ denotes the TD goal. For accurate and effective action execution, the BRMC system framework must take into account the robot's physical limitations as well as the real surroundings. Figure 7 depicts the BRMC system structure based on the enhanced QLA.

In the MC system framework, the environment sensing module is responsible for collecting real-time environment information, and the strategy learning module utilizes the improved QLA to perform self-learning and decision-making

based on the current state values to output the optimal strategy [30]. The action execution module is responsible for transforming the policy output from the learning module into actual robot actions. Moreover the error function is determined by Q-Learning, and its mathematical expression is shown in equation (12).

$$Loss = (r + \gamma \max_{a'} Q(s', a'; \theta') - Q(s, a; \theta))^2 \quad (12)$$

In equation (12), $Q(s, a; \theta)$ denotes the network output of the estimated value and $\max_{a'} Q(s', a'; \theta')$ denotes the output of the target network. The study employs a multi-modal data fusion approach. Firstly, it filters and calibrates the raw data from each sensor to eliminate noise and bias. Next, it extracts key features from the pre-processed data, such as target position, velocity, and the current pose of the robot. Then, it uses a Kalman filter to balance and fuse the features extracted from different sensors. Finally, it encodes the fused state estimation into a state vector as input for decision-making in the Q-learning algorithm.

IV. SIMULATION TEST OF MC-IBR MODEL WITH IMPROVED Q-LEARNING BASED ON MULTI-SENSOR DATA FUSION

In order to verify the effectiveness of the MC-IBR designed in this experiment, it is experimentally analyzed and explored. The analysis includes a detailed system parameterization to ensure precision and accuracy at each step. The main simulation software used is MATLAB R2023a, and its Reinforcement Learning Toolbox is used as an add-on for the Q-Learning implementation. Additionally, Webots R2023a is used as the robotic system modeling software to more accurately simulate the movements of the IBR. The computer configuration, however, is an Intel Core i9-11900K processor with an NVIDIA GeForce RTX 3080 graphics card, Dominator Platinum 32GB (2 × 16GB) DDR4 3200MHz memory, and Windows 10 Pro 64-bit operating system. Its sensor type, on the other hand, is a Hokuyo URG-04LX-UG01 laser range-finder. The experimental environment parameters are shown in Table 1.

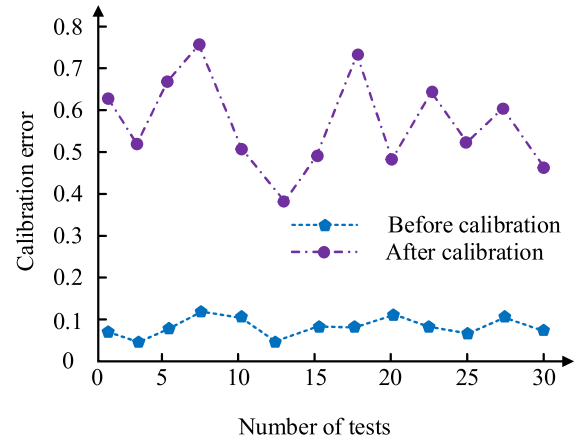
TABLE 1. Experimental environmental parameter.

Parameter Name	Description
Software	MATLAB R2023a, Webots R2023a, OpenCV 4.5.2
Hardware	Intel i9-11900K, NVIDIA RTX 3080, Corsair 32GB DDR4 3200MHz
Operating System	Windows 10 Pro
Sensor	Hokuyo URG-04LX-UG01 Laser Range Finder
Robot Model	KUKA youBot
Network Type	DQN with Experience Replay and Target Network

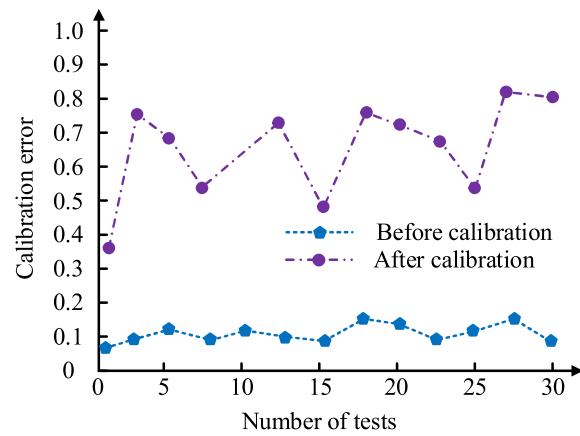
The experimental environment shown in Table 1 utilizes software tools such as MATLAB R2023a, Webots R2023a and OpenCV 4.5.2, and the operating system chosen is Windows 10 Pro to ensure the accuracy of the experiment. An existing state-of-the-art KUKA youBot robot model with a Hokuyo URG-04LX-UG01 laser rangefinder sensor is used. A uniform configuration is used aiming to achieve efficient and accurate robot MC. In basketball games, robots need to perform precise shooting and passing movements, which requires extremely high precision in posture control. In order to further verify the effectiveness of posture and distance calibration in practical applications, we applied this technology to basketball intelligent robots and tested it in real basketball game scenes. The effect of robot pose calibration in high-speed motion scenarios is shown in Figure 8.

In Figure 8, the performance of the robot is significantly improved after calibration at different angles. Specifically, in Figure 8(a), the highest value of error before calibration is 0.77 while the lowest value is 0.39 under the angle of traverse roll angle. Through accurate calibration, the highest and lowest values of error are reduced to 0.11 and 0.04, respectively, which verifies the importance and effectiveness of calibration. In Figure 8(b), for the calibration at pitch angle, the highest error before calibration reaches 0.86, and the lowest error is 0.37, which is a large error range that negatively affects the robot's operational accuracy. However, after calibration, the highest error is reduced to 0.15, and the lowest error is reduced to 0.06. It can be noted that the calibration of the robot at different angles can significantly reduce the error and improve the accuracy of its operation. The effect of robot distance calibration in the scene of rapid posture change is shown in Figure 9.

Figure 9 shows a significant improvement in distance error values between the robots after calibration at different angles. At the roll angle (Figure 9a), the robots' highest distance error before calibration is 1.26 and the lowest is 0.31. After calibration, the highest error is reduced to 0.62 and the lowest to 0.24. The data shows that calibration improves both the distance error between the robots and the operation accuracy. Before calibration, the highest distance error at the pitch angle is 1.43 and the lowest is 0.37, as shown in Figure 9(b).



(a) Effect of Basketball Robot Posture Calibration under Roll Angle



(b) Effect of Basketball Robot Posture Calibration under Pitch Angle

FIGURE 8. The effect of robot pose calibration in high-speed motion scenarios.

However, after calibration, the highest and lowest errors are reduced to 0.74 and 0.51, respectively. The calibration of the robots' distance effectively reduces errors and improves accuracy and synergy efficiency during their operations. Figure 10 shows the visualization results of the strategy function and action probability in different states.

In Figure 10(a), the improved QLA exhibits superior accuracy for robot MC in the stationary state. Compared with the traditional MC technique, its action probability is significantly reduced to 0.39, 0.43, 0.31, and 0.48, respectively. This accuracy makes the robot more stable and less susceptible to interference from external factors in the stationary state. Moreover in Figure 10(b), when the robot is in motion, the improved QLA still shows excellent MC accuracy. Its action probabilities are significantly higher compared to the traditional MC technique, which are 0.71, 0.65, 0.79, and 0.88, respectively. This improved accuracy allows the robot to perform tasks more flexibly and accurately in motion. The results of the comparison of the average MC return values of

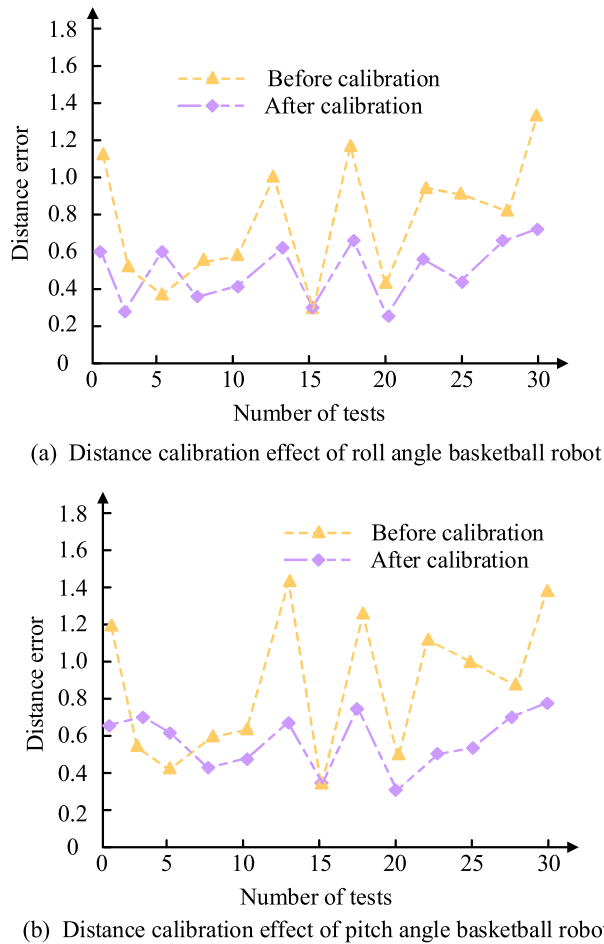


FIGURE 9. The effect of robot distance calibration in rapidly changing posture scenarios.

different algorithms under different competitions are shown in Figure 11.

Figure 11 demonstrates the significant difference in the mean return values produced by the different MC algorithms in the two different tournament environments (competitive and friendly tournaments). Figure 11(a) shows that the modified QLA used in the competitive match environment outperforms the traditional MC technique and the PID control technique in terms of mean return value of MC. The modified QLA achieves a mean return value of 412, while the traditional MC technique achieves the lowest mean return value of 168, and the PID control technique achieves a mean return value of 376. This demonstrates the clear advantage of the modified QLA in terms of the efficiency and effectiveness of MC. In Figure 11(b), the average MC return value of the improved QLA remains at the highest level of 149. the traditional MC technique has the lowest average return value of 23, and the PID control technique has the average MC return value of 136. The superiority of the improved QLA in terms of MC effectiveness is verified. Table 2 displays the results of performance comparisons for several MC algorithms.

Table 2 shows significant differences among the four algorithms in terms of recognition rate, response speed,

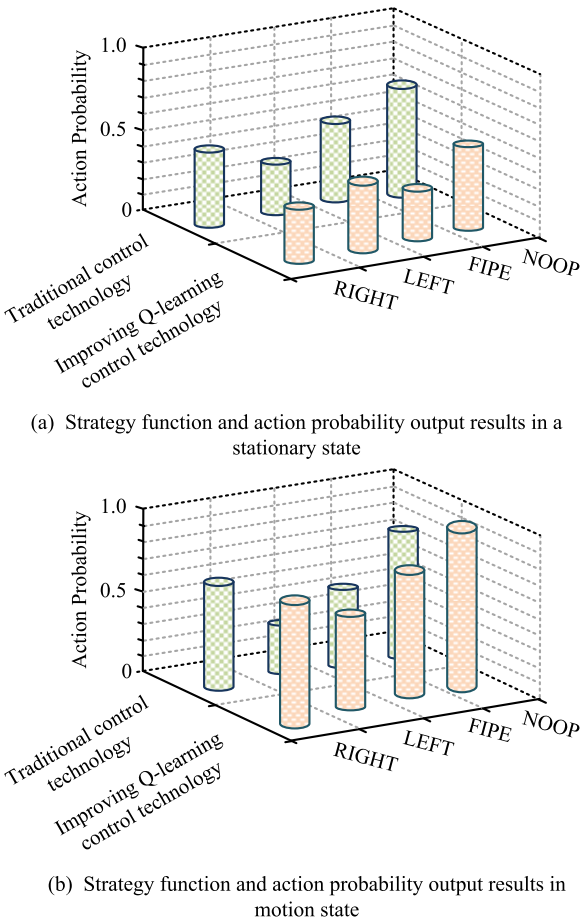
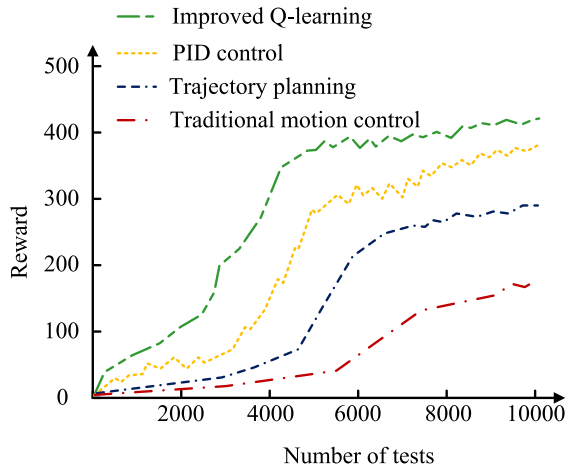


FIGURE 10. Visualization results of strategy functions and action probabilities in different states.

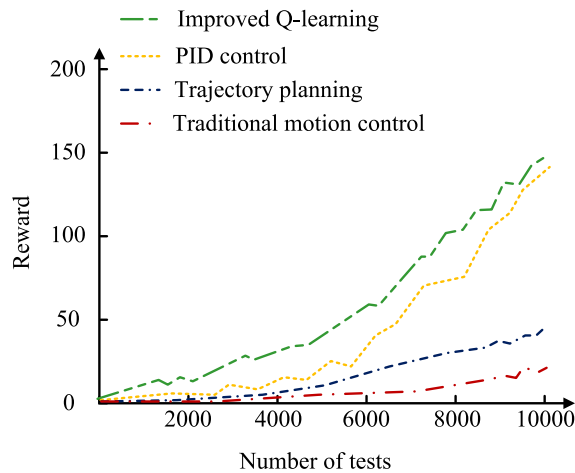
TABLE 2. Performance comparison results of different action control algorithms.

Algorithm	Recognitio n Rate	Response speed	Performance stability	Error rate
Improved Q-learning algorithm	99.35	67.85	97.55	7.65
PID control algorithm	98.74	58.27	93.68	20.45
Trajectory planning algorithm	90.66	45.31	87.25	28.78
Traditional control algorithms	86.19	41.17	83.64	36.62
Multi sensor data fusion technology [31]	91.36	53.15	91.06	20.02
Visual perception basketball shooting direction correction method [32]	93.74	55.37	91.33	19.61

performance stability, and error rate. The improved QLA algorithm performs well in all four indicators, with a recognition rate of 99.35, response speed of 67.85, performance stability of 97.55, and error rate of 7.65. The PID control algorithm performs at 98.74, 58.27, 93.68, and 20.45 in the same indicators, respectively. The traditional



(a) Comparison of the Average Return Value of Action Control for Basketball Robots Using Different Algorithms in Competitive Games

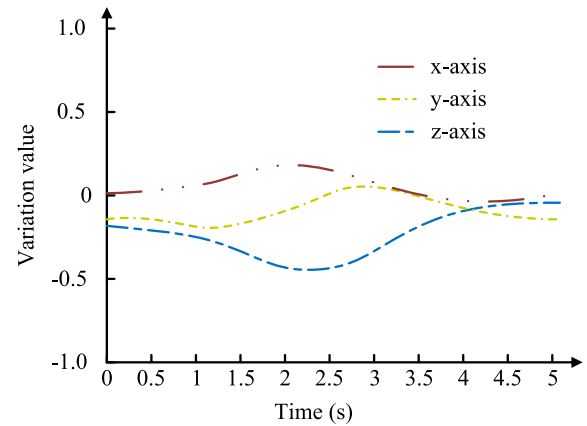


(b) Comparison of the Average Return Value of Action Control for Basketball Robots Using Different Algorithms in Friendship Tournaments

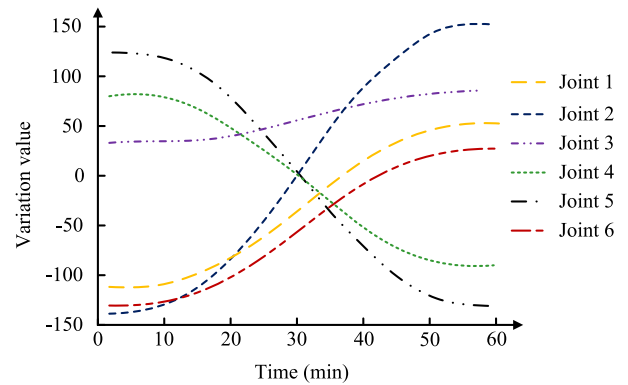
FIGURE 11. Comparison results of average return value of action control for different algorithms in different competitions.

control algorithm performs at 86.19, 41.17, 83.64, and 36.62, respectively. The study by F Shi et al. evaluates the performance of basketball robots using MSDF technology. The results show scores of 91.36, 53.15, 91.06, and 20.02 for various indicators. Huang et al. proposed a visual perception shooting direction correction method with a recognition rate of 93.74, performance stability of 91.33, and an error rate of 19.61. It can be noted that the improved QLA outperforms the remaining three algorithms in all the evaluation indexes, demonstrating high efficiency and stability in the robot control model. The results of the joint space planning robot arm end-points over time and angle are shown in Figure 12.

In Figure 12(a), the trajectory component change trend of the joint end-point of the robotic arm of the basketball robot in 3D space is smooth. Its trajectory component changes smoothly without obvious abrupt changes, which verifies



(a) The results of joint planning robot arm end changes over time



(b) Joint space planning: Mechanical joint angles change over time

FIGURE 12. Joint space planning robot arm end-point addition and angle change results at any time.

that the changes of the joint end-points of the robotic arm are continuous and smooth during its movement. Its values produced at the 5ths in different axis directions are 0.12, -0.16 and -0.09 , respectively, reflecting the fine capture and control of the movement changes of the robotic arm during the movement. In Figure 12(b), the trajectories of the different robotic arm end-points are not a straight line, which shows that the process of changing the joint angle is smooth and there is no sudden and sharp change. The robotic arm of the basketball robot is able to capture and control the movement changes finely during the movement process, and the changes of its joint angles are continuous and smooth. The results of the trajectory test error of the mechanical leg and hand of the basketball robot are shown in Figure 13.

In Figure 13, there is a significant difference between the leg and hand trajectory test results and the errors they produce as the number of tests increases. In Figure 13 (a), the error value is the largest at 100 times of testing, reaching 0.27. However, when the number of tests is increased to 500 times, the error value decreases to a minimum of 0.16. The trajectory test results of the legs of the basketball robot are getting closer and closer to the real value as the number of tests is increased, and the error gradually decreases. In Figure 13(b), the error value is the largest at 200 times of testing, which is 0.41.

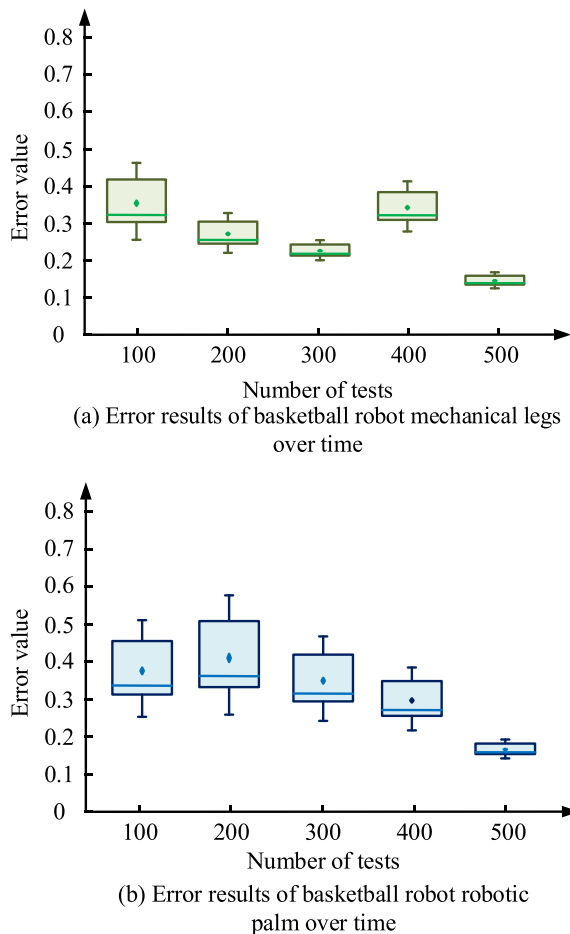


FIGURE 13. Error results of trajectory testing for the mechanical legs and palms of basketball robots.

When the number of tests is increased to 500 times, the error value is reduced to the minimum, which is 0.18. The accuracy of the trajectory test results of the hand also increases with the increase in the number of tests. The accuracy of the trajectory test results for both the leg and the palm of the basketball robot increased and the error gradually decreased.

V. CONCLUSION

With the rapid development of artificial intelligence and robotics technology, the application of intelligent robots in various fields is becoming increasingly widespread, among which the MC technology of basketball intelligent robots has attracted people's attention. However, due to the complexity of the environment and the robot's physical characteristics, improving the accuracy and stability of its MC remained a challenge. This research proposed a study of MC-IBR technology based on improved Q-Learning under MSDF. The experimental results indicated that the traverse angle had a range of error values before calibration, with the highest being 0.77 and the lowest being 0.39. However, after precise calibration, the range of error values significantly

decreased, with the highest being 0.11 and the lowest being 0.04. Similarly, the pitch angle had a highest error value of 0.86 and a lowest error value of 0.37 before calibration. After calibration, the highest and lowest errors were reduced to 0.15 and 0.06, respectively. In the competition environment, the QLA outperformed the conventional MC technique and the PID control technique with a mean return value of 412, compared to 168 and 376, respectively. The proposed technology can significantly reduce control errors in robot actions and achieve higher average return values in competitive environments. This provides technical support and research direction for the field of robot MC and opens up new avenues for the application of robot technology in sports such as basketball. However, the study did not consider more complex practical application environments or other factors that may affect control effectiveness. For example, basketball intelligent robots face more complex lighting and noise interference in actual games. Additionally, the interactive dynamics of robots cooperating with human players or other robots in practical applications may also impact control effectiveness. Therefore, future research should conduct in-depth studies on the robustness of improved Q-learning algorithms in more complex environments and their uncertainty in practical applications. Additionally, research should explore the integration of algorithms with on-site sensor networks, decision-making, and collaborative work mechanisms for multi-robot collaboration.

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